VerifactAI — Misinformation Detector

Simple Project Review & Presentation Guide

What is VerifactAI? (Simple Explanation):
VerifactAI is an intelligent tool that reads any news article or social media post and checks if it's telling the truth. It ignores personal opinions, picks out real factual claims, searches the live internet for proof, compares the claims against real web evidence using AI, and gives a clear **Risk Score (0 to 100)** and **Risk Level (Low, Medium, High)**.

*This guide explains **HOW** each part works and **WHY** we built it this way in plain, simple English.*

1. The Big Picture: How VerifactAI Works

VerifactAI works like a team of human fact-checkers, broken into 3 simple steps:

Step 0: Input Text & Quality Check
Reads the whole article and checks reading difficulty and how smoothly sentences connect.
Step 1: Fact Finder & Opinion Filter • Splits the text into individual sentences. • Uses a smart AI model to keep ONLY factual statements (filters out personal opinions). • Checks if the text looks like it was written by ChatGPT.
■ *Smart Shortcut: If the text is 100% opinion or chatter, it stops early to save time and money.*
Step 2: Live Internet Search • Searches the live web for each factual claim using Tavily Search. • Picks the 3 most relevant web page paragraphs as evidence.
Step 3: Truth Checker & Risk Calculator • Uses an AI reasoning model to check if the web evidence supports or contradicts the claim. • Calculates a final Risk Score (0 to 100) and assigns a Risk Level (Low, Medium, High).
Final Output: Simple Summary Card & Risk Report

2. Step-by-Step Breakdown: HOW and WHY

Pre-Check: Writing Style & Reading Difficulty

HOW IT WORKS: Before checking facts, VerifactAI measures how easy or hard the text is to read (reading grade level) and how smoothly ideas flow from one sentence to the next (sentence coherence).

WHY WE DID THIS: Fake news and AI-generated articles often feel 'off'. Sensational clickbait is usually written at a very basic level to grab attention, while fake AI articles often have choppy, disconnected sentences. Measuring style helps spot weird writing patterns.

Step 1: Fact Finder & Opinion Filter

HOW IT WORKS:

1. Splits the article into sentences.
2. Passes each sentence through a specialized AI model (DeBERTa) that scores whether the sentence is a **verifiable fact** or just a **personal opinion**.
3. Passes sentences through an AI-detector model (RoBERTa) to calculate the chance it was generated by ChatGPT.
4. Only sentences that are strong factual claims (score ≥ 0.65) move to the next step.

Design Decision	Why It Matters (In Plain English)
Batch Processing	We feed all sentences into the AI model at once instead of one by one. This makes the system run up to 5x faster.
Filtering Out Opinions	You can't fact-check feelings like *'I think rainy days are sad'*. Filtering out opinions ensures we only waste web search effort on real facts.
Early Exit Guard	If an article has ZERO factual claims (e.g. casual chatter), the system stops immediately. This saves 100% of web search fees and API costs.

Step 2: Live Internet Evidence Search

HOW IT WORKS:

1. For every factual claim, VerifactAI performs a live web search using Tavily Search API.
2. It downloads the text of top web pages and uses a lightweight AI model (MiniLM) to rank the snippets by how closely they match the meaning of the claim.
3. It keeps the **top 3 best matching snippets** as proof.

Design Decision	Why It Matters (In Plain English)
Tavily Search API	Standard Google searches just give short titles. Tavily pulls full, clean text from web pages so our system gets real context.
Smart Relevance Ranking	Search engines rank pages by popularity. Our local AI model ranks snippets specifically by how directly they talk about the claim.
Keeping Top 3 Snippets	Focusing on the 3 best snippets removes clutter and keeps the verification fast and low-cost.

Step 3: Truth Verification & Risk Calculation

HOW IT WORKS:

1. **AI Reasoning:** Sends the claim and top 3 web snippets to an AI model (Groq LLM). The AI is strictly instructed: *'Reason ONLY from the provided web proof. If evidence is missing, say insufficient evidence.'*
2. **Verdict:** The AI outputs one of three verdicts: **Supported** (True), **Contradicted** (False), or **Insufficient Evidence**.
3. **Simple Risk Score Formula:**
   Risk Score = 100 × [(0.05 × AI-Text Score) + (0.05 × Fact Score) + (0.90 × Contradiction Score)]
4. **Risk Levels:** 0–33 = Low Risk (Green) | 34–66 = Medium Risk (Yellow) | 67–100 = High Risk (Red). *(If evidence is missing, the risk is capped at Medium so we don't falsely accuse someone).*

Design Decision	Why It Matters (In Plain English)

Strict AI Instructions	Forcing the AI to look ONLY at web proof stops it from guessing or making up fake information (hallucinating).
'Insufficient Evidence' Option	If web search finds no clear proof, admitting 'we don't know' is much safer than making a false guess.
Fixed Math Formula	Instead of letting the AI guess a random risk number, we use a simple math formula that is 100% transparent and reliable.
Heavy Contradiction Weight (0.90)	Our tests proved that whether real web evidence contradicts a claim is by far the most important factor in catching fake news.

3. How We Tested & Improved the System

We didn't just write code — we tested it on real claims to make sure it works accurately.

1. Formula Tuning (calibrate_weights.py):
• We tested 24 benchmark claims (True facts, False myths, and Unverifiable rumors).
• We tried hundreds of weight combinations in a grid search.
• Tuning the weights boosted our classification accuracy from **66.67%** up to **75.00%**!

2. Real Academic Benchmark (evaluate.py):
• We evaluated the pipeline against 60 human-checked claims from the famous **FEVER** fact-checking dataset.
• We added automatic saving so if the internet drops, the test resumes right where it left off without wasting API calls.

4. Backend API & Frontend Dashboard

VerifactAI comes with a clean web server (api.py using FastAPI) and a simple test webpage (index.html).

Component	What It Does (Simple Explanation)
POST /analyze	The main endpoint. Send any article text, and it returns the risk score, claims, web sources, and time taken.
index.html	A visual dashboard where you can paste text, click 'Analyze', and instantly see colored risk cards and claim breakdowns.
Time Tracking	Tracks how many seconds each step took (Step 1, Step 2, Step 3) so you can see where time was spent.

5. Simple Presentation Q&A; (Review Cheat Sheet)

Here are 5 easy answers to common questions reviewers might ask tomorrow:

Q1: Why not just ask ChatGPT if an article is true?

A1: ChatGPT relies on old memory and can easily hallucinate (make up fake facts). VerifactAI searches the live internet for real proof and forces the AI to look ONLY at that proof.

Q2: What is the purpose of the 'Early Exit' in Step 1?

A2: If an article is 100% opinions or personal banter, doing web searches and AI queries is a waste of time and money. Early exit stops the system immediately and returns 0 risk.

Q3: How did you pick the numbers in your Risk Formula?

A3: We didn't guess. We tested hundreds of combinations on 24 real benchmark claims. Giving 90% weight to evidence contradiction gave the highest accuracy (75%).

Q4: What if internet search returns no results?

A4: The system marks the claim as 'Insufficient Evidence' and caps the risk tier at Medium. It never makes a wild guess when evidence is missing.

Q5: What models are you running locally vs on the cloud?

A5: We run DeBERTa (fact-filter) and MiniLM (similarity ranker) locally on our computer for speed. We only call cloud APIs for web search (Tavily) and final reasoning (Groq).